

Kidney Transplant — CMV Risk, Timing & Clinical Spectrum (CMV 1)

THE CMV REACTIVATION LAZARETTO — POST-TRANSPLANT CMV: RISK, TIMING & CLINICAL SPECTRUM

1-6 MONTHS post-transplant. D+/R- = highest risk in SOT. Fever + leukopenia + colitis = the classic syndrome.



1-6 month window, D+/R- in SOT. Fever + leukopenia + transaminitis = CMV syndrome. GI is most common tissue disease in SOT; pneumonitis most feared in HCT. CMV worsens rejection and graft outcomes.

1. 1-6 month window

WHAT YOU SEE

Banner: '1-6 months post-transplant, D+/R- = highest risk'

WHAT IT ENCODES

CMV is the prototype opportunistic infection of the 1-6 month post-transplant window, the period of maximal net immunosuppression after induction has been given but before doses are tapered. Without prophylaxis, CMV disease peaks here; universal valganciclovir prophylaxis (typically 3-6 months, longer in D+/R-) shifts disease into a 'late-onset' pattern that appears AFTER prophylaxis stops. Mnemonic: 1-6 months = 'opportunistic middle window' (CMV, PJP, BK, fungal, reactivation TB).

BOARD TRAP

Board trap: a stem placing CMV at <1 month is wrong — that window is nosocomial/surgical/donor-derived (line sepsis, wound, C. diff, aspiration). CMV >6 months is possible only as 'late-onset' after prophylaxis ends; the classic, default association is the 1-6 month window.

2. SOT — solid organ

WHAT YOU SEE

A solid-organ recipient figure clutching a transplanted kidney — in SOT the donated organ itself carries latent CMV, so DONOR serostatus drives risk

WHAT IT ENCODES

In solid organ transplant (SOT), the transplanted allograft itself carries latent CMV in donor endothelial and dendritic cells, so a seronegative recipient receiving a seropositive organ is exposed to primary infection. This is why donor serostatus dominates SOT risk stratification, and why the kidney/liver/lung/heart graft is the vehicle. Net state of immunosuppression (induction agent, rejection treatment) modulates how much that exposure translates into disease.

BOARD TRAP

Board trap: don't apply the HCT (bone marrow) risk logic to SOT. In SOT the donated solid organ transmits virus, so D+/R- is worst; assuming 'recipient seropositivity is the main risk' is the HCT rule, not the SOT rule.

3. D+/R- highest risk

WHAT YOU SEE

Donor-positive / recipient-negative pairing flagged highest

WHAT IT ENCODES

CMV-seropositive donor into a CMV-naive (seronegative) recipient — D+/R- — is the single highest-risk serostatus combination in SOT because the recipient has no pre-existing CMV-specific T-cell immunity to contain primary infection from the graft. These patients get the longest prophylaxis (valganciclovir 900 mg daily, renally dosed, often 6 months) and the most intensive monitoring. Risk gradient: D+/R- > D+/R+ ≈ D-/R+ > D-/R- (lowest).

BOARD TRAP

Board trap: D+/R+ or D-/R+ are NOT the highest risk — the answer is D+/R- (donor positive, recipient negative). Picking R+ confuses SOT with HCT, where recipient seropositivity (R+) drives reactivation risk.

4. HCT recipient

WHAT YOU SEE

A stem-cell/HCT recipient figure with an IV bag of marrow — here the patient's OWN ablated immunity means RECIPIENT seropositivity (R+) drives reactivation

WHAT IT ENCODES

In hematopoietic cell transplant (HCT/bone marrow), the patient's own immune system is ablated and replaced by donor marrow, so it is the RECIPIENT'S latent CMV (R+) that reactivates while donor-derived immunity has not yet reconstituted. Therefore R+ is the high-risk serostatus, and a seropositive donor can actually be protective by transferring CMV-specific immunity. This is the mirror image of the SOT rule.

BOARD TRAP

Board trap: applying D+/R- logic to a bone-marrow transplant stem is wrong. SOT: donor matters most (D+/R- worst). HCT: recipient seropositivity (R+) matters most; here D+ donor may even help.

5. Latency crypt / reactivation

WHAT YOU SEE

Latent CMV in endothelial & dendritic cells reactivating when immunity drops

WHAT IT ENCODES

CMV establishes lifelong latency in myeloid progenitors, vascular endothelial cells, and dendritic cells; it is contained by CMV-specific CD8 T cells. When immunosuppression (especially T-cell-depleting agents like ATG/alemtuzumab, or rejection treatment) lowers this surveillance, the virus reactivates. This explains both reactivation in R+ patients and why pulse anti-rejection therapy is a discrete CMV trigger even late post-transplant.

BOARD TRAP

Board trap: a patient treated for acute rejection with ATG who then develops fever/cytopenias months later is NOT having 'recurrent rejection' by default — the added T-cell depletion is a classic CMV reactivation trigger; check a CMV PCR before escalating immunosuppression further.

6. Asymptomatic viremia

WHAT YOU SEE

Sleeping patient, PCR+ no symptoms

WHAT IT ENCODES

Asymptomatic CMV viremia = detectable CMV by quantitative PCR (reported in IU/mL) with no clinical signs. It is the trigger for the 'preemptive' strategy: instead of giving everyone prophylaxis, high-risk patients are monitored with serial PCR and treated with valganciclovir only once viral load crosses a threshold or is rising. Quantitative PCR is also how response to therapy is tracked (treat until 1-2 consecutive negative/undetectable results).

BOARD TRAP

Board trap: in the preemptive-monitoring strategy, a rising/positive PCR in an asymptomatic patient still warrants treatment — answering 'observe because no symptoms' is wrong. Conversely, CMV IgG/IgM serology is for pre-transplant risk stratification, NOT for diagnosing active disease (PCR is the answer).

7. CMV syndrome

WHAT YOU SEE

A feverish patient labelled with fever + leukopenia + transaminitis — CMV viremia with systemic features but NO biopsy-proven organ invasion

WHAT IT ENCODES

CMV syndrome = CMV viremia plus systemic features (fever, malaise, leukopenia ± thrombocytopenia, mild transaminitis/atypical lymphocytosis) WITHOUT biopsy-proven end-organ tissue invasion. It is the milder end of symptomatic disease and overlaps with a mononucleosis-like picture. Treatment is IV ganciclovir 5 mg/kg q12h or oral valganciclovir 900 mg q12h (renally dosed) plus reduction of immunosuppression, continued until viral clearance.

BOARD TRAP

Board trap: fever + leukopenia + transaminitis in the 1-6 month window is CMV syndrome, not simply 'drug-induced cytopenia from MMF' or 'allograft rejection.' The discriminator is a positive CMV PCR; missing this and merely cutting MMF leaves the CMV untreated.

8. Tissue-invasive disease

WHAT YOU SEE

The sickest patient in the 'escalation' rooms, marked tissue-invasive — biopsy-proven end-organ CMV (colitis, pneumonitis, hepatitis, retinitis)

WHAT IT ENCODES

Tissue-invasive (end-organ) CMV means demonstrable organ involvement: colitis, pneumonitis, hepatitis, esophagitis, nephritis, or retinitis, often with biopsy showing characteristic 'owl's eye' intranuclear inclusions. It is more severe than CMV syndrome and usually requires IV ganciclovir first-line plus reduction of immunosuppression; ganciclovir-resistant disease (UL97/UL54 mutations) is treated with foscarnet, cidofovir, or letermovir.

BOARD TRAP

Board trap: a CMV-positive patient with colitis/pneumonitis needs IV ganciclovir, not just oral valganciclovir or observation; treating tissue-invasive disease like mild syndrome under-treats it. Persistent disease despite adequate ganciclovir = suspect resistance → switch to foscarnet, not 'more ganciclovir.'

9. GI / colitis most common

WHAT YOU SEE

Tissue-invasive organ wing: GI/colitis flagged 'most common'

WHAT IT ENCODES

Gastrointestinal disease — especially colitis (bloody diarrhea, abdominal pain) but also esophagitis/gastritis — is the most common tissue-invasive manifestation of CMV in solid organ transplant. Diagnosis is by endoscopic biopsy showing CMV inclusions/immunohistochemistry; note that CMV PCR in blood can be negative in localized GI disease, so a negative serum PCR does not exclude colitis.

BOARD TRAP

Board trap: a SOT patient with bloody diarrhea and a NEGATIVE serum CMV PCR can still have CMV colitis — the next step is colonoscopy with biopsy, not ruling CMV out. Also, GI colitis (SOT) versus pneumonitis (most feared in HCT) is a commonly swapped pair.

10. Pneumonitis (HCT feared)

WHAT YOU SEE

Lung icon, 'most feared in HCT'

WHAT IT ENCODES

CMV pneumonitis (fever, dry cough, hypoxia, diffuse interstitial infiltrates) is the most feared and historically most lethal tissue-invasive form in HCT recipients, with high mortality if untreated. Management combines IV ganciclovir with IVIG in HCT pneumonitis. It must be distinguished from the other major 1-6 month pulmonary opportunist, PJP, which prophylaxis (TMP-SMX) normally prevents.

BOARD TRAP

Board trap: hypoxic interstitial pneumonia in the opportunistic window has two big answers — CMV and PJP. The discriminator: PJP gives markedly elevated LDH and is prevented by TMP-SMX; if the patient is on TMP-SMX and CMV-mismatched/off CMV prophylaxis, lean CMV and confirm with BAL.

11. Retinitis (late/AIDS-like)

WHAT YOU SEE

Eye icon, 'usually late'

WHAT IT ENCODES

CMV retinitis (floaters, painless visual field loss, 'pizza-pie' hemorrhagic exudates on fundoscopy) reflects profound, prolonged immunosuppression and tends to occur later than other tissue-invasive forms. It is classic in advanced AIDS (CD4 <50) but also occurs in heavily immunosuppressed transplant patients; it requires urgent ophthalmology and systemic ± intravitreal ganciclovir/valganciclovir.

BOARD TRAP

Board trap: painless visual loss with 'pizza-pie' retinal findings is CMV retinitis — don't anchor on HSV/VZV acute retinal necrosis (which is rapidly progressive, painful, peripheral) or toxoplasma chorioretinitis. CMV retinitis tends to be late and demands systemic antiviral, not topical steroids.

12. Indirect effects

WHAT YOU SEE

Cloud: acute rejection, other opportunistic infections, mortality

WHAT IT ENCODES

Beyond direct organ damage, CMV exerts immunomodulatory 'indirect effects': it upregulates alloimmunity (increasing acute and chronic allograft rejection and graft loss), predisposes to OTHER opportunistic infections and post-transplant lymphoproliferative disorder, and is associated with increased overall mortality and accelerated atherosclerosis/vasculopathy. This is the core rationale for routine CMV prophylaxis even though many infections would be 'mild.'

BOARD TRAP

Board trap: the reason for universal CMV prophylaxis is not just preventing colitis — the indirect effects (more rejection, more superinfection, higher mortality) are the high-yield justification. A question implying CMV 'only matters if symptomatic' is wrong.

13. Timeline: bacterial→opportunistic→community

WHAT YOU SEE

Hallway timeline <1mo bacterial/surgical, 1-6mo CMV/PJP/fungal, >6mo community

WHAT IT ENCODES

The classic three-phase timeline: <1 month = nosocomial/surgical/donor-derived (wound infection, line sepsis, UTI, aspiration pneumonia, C. diff, donor-transmitted infection); 1-6 months = opportunistic on a background of maximal immunosuppression (CMV, PJP, BK virus, fungal/Aspergillus, Nocardia, reactivation TB/Strongyloides) — these are SUPPRESSED by prophylaxis (TMP-SMX, valganciclovir); >6 months = community-acquired (CAP, influenza, UTI) plus late opportunistic/disease in those still heavily immunosuppressed.

BOARD TRAP

Board trap: matching the wrong bug to the window. PJP/CMV at week 1 is wrong (too early — that's surgical/nosocomial). Community-acquired pneumonia at month 1 is wrong (that's the >6 month phase). Remember prophylaxis shifts opportunistic infections later, so 'late' CMV/PJP appears just after prophylaxis stops.

14. Boards trap: R- not on prophylaxis

WHAT YOU SEE

Card: 'CMV+ in transplant + no prophylaxis → PCR'

WHAT IT ENCODES

High-yield reflex: any transplant patient in the opportunistic window who is NOT on (or has just finished) CMV prophylaxis and presents with fever, leukopenia/thrombocytopenia, transaminitis, or GI symptoms → send a quantitative CMV PCR. This is especially true for D+/R- recipients and patients recently treated for rejection with ATG. Confirmatory and monitoring tool is quantitative PCR; serology is only for pre-transplant risk.

BOARD TRAP

Board trap: the wrong first move is ordering CMV IgM/IgG serology to diagnose active disease, or empirically increasing immunosuppression for presumed rejection. The right answer is quantitative CMV PCR; serology cannot distinguish active disease from prior exposure in the immunosuppressed.